STPM 2024 PHYSICS 960/3

Questions and Answers

Values of constants: $h = 6.63 \times 10^{-34}$ *J s,* $c = 3.00 \times 10^{8}$ *m* s^{-1}, $e = 1.60 \times 10^{-19}$ *C,* $m_e = 9.11 \times 10^{-31}$ *kg,* $g = 9.81$ *m* s^{-2}.

Section A [15 marks]

1. Which is true about the oscillation system that experiences critical damping?

A Oscillates with constant amplitude
B Oscillates with decreasing amplitude
C Does not return to the equilibrium position
D Returns to the equilibrium position without oscillation

Answer: D

A critically damped system returns to the equilibrium position in the shortest time without oscillating.

Explanation: Underdamped systems oscillate with decreasing amplitude; undamped systems keep a constant amplitude.

2. When a stretched wire between two fixed points is plucked, a standing wave is produced. Which statement is **not** true about the standing wave?

A The speed of the standing wave is zero.
B All the particles on the wire have the same amplitude.
C All the particles between two consecutive nodes oscillate at the same frequency.
D The wavelength of the standing wave equals the distance between three consecutive antinodes.

Answer: B

In a standing wave the amplitude varies along the wire: zero at the nodes and maximum at the antinodes. Statement B is not true.

Explanation: A standing wave does not travel (zero speed of energy transfer), all points between two nodes oscillate in phase at the same frequency, and one wavelength spans three consecutive antinodes.

3. The magnetic vibrations of an electromagnetic wave is given by $B = B_o \sin(6 \times 10^8 \pi t - 2\pi x)$, where t is in seconds and x is in metres. The electromagnetic wave is representing

A an X-ray B an ultraviolet C a radio wave D a visible light

Answer: C

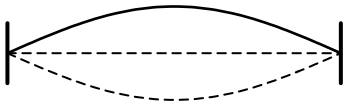
$$\omega = 6 \times 10^8 \pi \Longrightarrow f = 3 \times 10^8 \text{ Hz}; \; k = 2\pi \Longrightarrow \lambda = 1 \text{ m}$$

A wavelength of about 1 m lies in the radio-wave region.

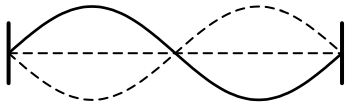
Explanation: Check: $v = f\lambda = 3 \times 10^8$ m s^{-1}, the speed of light. Visible light has $\lambda \approx 10^{-7}$ m and X-rays $\approx 10^{-10}$ m.

4. An elastic string is stretched between two points and then is plucked at its mid-point. Which diagram shows the mode of vibration of the string?

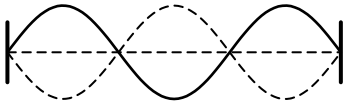
A



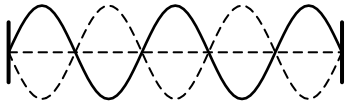
B



C



D

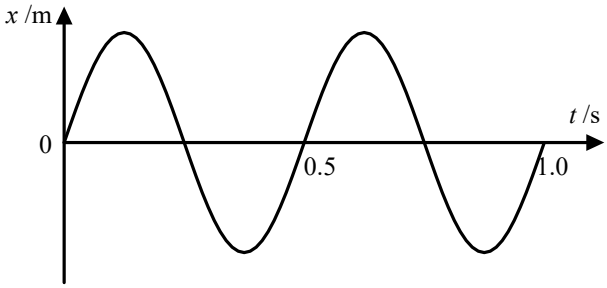


Answer: A

Plucking at the mid-point makes an antinode there, with nodes at both fixed ends: the fundamental mode (one loop).

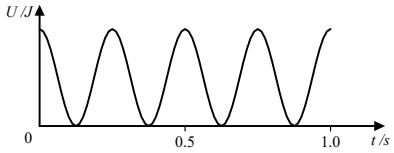
Explanation: Modes with a node at the mid-point (B and D) cannot be excited strongly by plucking at the centre.

5. The variation of displacement, x, with time, t, for a particle that moves in simple harmonic motion is shown in the graph below.

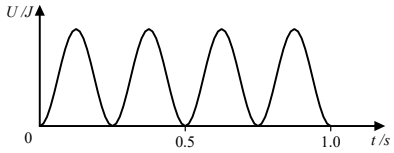


Which graph represents the variation of potential energy, U, with time, t, for the above situation?

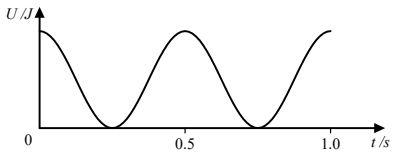
A



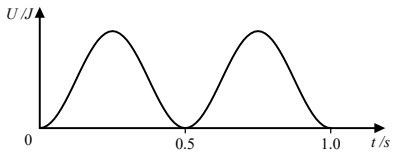
B



C



D



Answer: B

$x = A \sin \omega t$ with period 0.5 s; $U = \frac{1}{2}kx^2 \propto \sin^2 \omega t$.

U starts at zero and has period 0.25 s (half the period of x): graph B.

Explanation: Potential energy is maximum at each extreme position (twice per oscillation) and zero at each pass through equilibrium.

6. If the sound intensity level increases by 30 dB, what is the value of $\frac{\text{final sound intensity}}{\text{initial sound intensity}}$?

A 3 B 30 C 48 D 1000

Answer: D

$$\Delta\beta = 10 \log \frac{I_2}{I_1} = 30 \Longrightarrow \frac{I_2}{I_1} = 10^3 = 1000$$

Explanation: Each 10 dB increase multiplies the intensity by 10.

7. An object is placed in front of a concave mirror with the radius of curvature 10 cm. An inverted image is formed at 700 cm in front of the mirror. If the size of the image is 55 cm, what is the size of the object?

A 0.40 cm B 0.51 cm C 0.80 cm D 5.0 cm

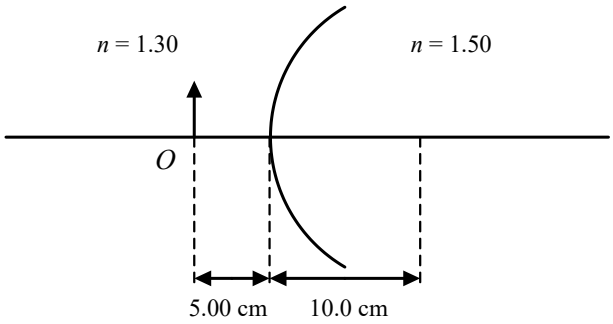
Answer: A

$$f = \frac{r}{2} = 5.0 \text{ cm}; \frac{1}{u} = \frac{1}{5.0} - \frac{1}{700} \Longrightarrow u = 5.04 \text{ cm}$$

$$m = \frac{v}{u} = \frac{700}{5.04} = 139 \Longrightarrow \text{object size} = \frac{55}{139} = 0.40 \text{ cm}$$

Explanation: An inverted image in front of the mirror is real, so $v = +700$ cm. The object is just beyond the focal point, giving a large magnification.

8. An object is placed at point *O* in a medium of refractive index 1.30 and 5.00 cm in front of a convex spherical surface of a block as shown in the diagram below.



If the refractive index of the block is 1.50 and the radius of curvature of the spherical surface is 10.0 cm, where is the position of the image formed?

A 1.25 cm on the left of the object
B 6.25 cm on the left of the object
C 6.25 cm on the right of the object
D 11.3 cm on the right of the object

Answer: A

$$\frac{n_1}{u} + \frac{n_2}{v} = \frac{n_2 - n_1}{r} : \frac{1.30}{5.00} + \frac{1.50}{v} = \frac{1.50 - 1.30}{10.0}$$

$v = -6.25$ cm: a virtual image 6.25 cm to the left of the surface, i.e. 1.25 cm to the left of the object

Explanation: The surface is convex towards the object, so $r = +10.0$ cm. A negative image distance means the image is on the same side as the object.

9. Which statement is **not** true about Huygens's principle?

A The secondary wavelets are spherical waves.
B The new wavefront is the tangent to all the secondary wavelets.
C Each point on a primary wavefront serves as a secondary point source.
D The distance between two adjacent wavefronts is equal to two wavelengths.

Answer: D

Adjacent wavefronts (e.g. crest to crest) are one wavelength apart, not two. Statement D is not true.

Explanation: Each point on a wavefront acts as a source of spherical secondary wavelets, and the new wavefront is their common tangent.

10. A double slit is separated by a distance 1.4 mm. The double slit is illuminated by monochromatic light of wavelength 460 nm. The interference pattern is observed on a screen 3.0 m away from the slit. What is the distance between the 3rd dark fringe and the 9th bright fringe?

A 6.4 mm B 5.9 mm C 4.4 mm D 3.0 mm

Answer: A

$$\text{Fringe separation } \Delta y = \frac{\lambda D}{d} = \frac{(460 \times 10^{-9})(3.0)}{1.4 \times 10^{-3}} = 0.986 \text{ mm}$$

3rd dark fringe at $2.5\Delta y$, 9th bright fringe at $9\Delta y$: distance = $6.5\Delta y$ = 6.4 mm

Explanation: Counting from the central bright fringe: the nth dark fringe is at $(n - ½)\Delta y$ and the nth bright fringe at $n\Delta y$ (both on the same side).

11. Photoelectrons are emitted when light is incident onto a metal surface. Which is true about the photoelectrons if the light is replaced with a shorter wavelength of light but with the same intensity?

A The rate of emission of the photoelectrons increases.
B The rate of emission of the photoelectrons decreases.
C The maximum kinetic energy of the photoelectrons increases.
D The maximum kinetic energy of the photoelectrons decreases.

Answer: C

A shorter wavelength means a higher photon energy, so $K_{max} = hf - W$ increases.

Explanation: With the same intensity, the higher-energy photons arrive at a lower rate, so the emission rate would not increase.

12. An electron, P, has kinetic energy, E_P, while another electron, Q, has kinetic energy, E_Q. What is the ratio of the de Broglie wavelength of P to that of Q?

A $\frac{E_Q}{E_P}$ B $\frac{E_P}{E_Q}$ C $\sqrt{\frac{E_Q}{E_P}}$ D $\sqrt{\frac{E_P}{E_Q}}$

Answer: C

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{(2mE)}} \Longrightarrow \frac{\lambda_P}{\lambda_Q} = \sqrt{(\frac{E_Q}{E_P})}$$

Explanation: Both are electrons (same mass), so the wavelength is inversely proportional to the square root of the kinetic energy.

13. The operation of an X-ray tube requires a voltage of 75 kV. What is the momentum of the electron in order to reach the target?

A 1.0×10^{-22} kg m s^{-1}
B 1.5×10^{-22} kg m s^{-1}
C 4.7×10^{-24} kg m s^{-1}
D 2.2×10^{-44} kg m s^{-1}

Answer: B

$$p = \sqrt{(2meV)} = \sqrt{(2(9.11 \times 10^{-31})(1.60 \times 10^{-19})(75 \times 10^{3}))} = 1.5 \times 10^{-22} \text{ kg m s}^{-1}$$

Explanation: The kinetic energy gained is eV, and $p = \sqrt{(2mK)}$. (Relativistic effects are ignored.)

14. Which is true about nuclear binding energy?

A The energy is required to combine the nucleons to form a nucleus.
B The energy is released when a heavy nucleus splits into two lighter nuclei.
C The energy is released when two light nuclei combine to form a heavier nucleus.
D The energy is required to separate a nucleus completely into individual nucleons.

Answer: D

Binding energy is the energy required to separate a nucleus completely into its individual protons and neutrons.

Explanation: It equals the energy released when the nucleus is formed from its nucleons, Δmc^2.

15. When a nucleus $^{235}_{92}\text{U}$ is bombarded by a thermal neutron $^{1}_{0}\text{n}$, nuclear fission occurs. One of the products is nucleus $^{99}_{40}\text{Zr}$. The other products are

A $^{137}_{50}\text{Sn}$ and two neutrons
B $^{134}_{50}\text{Sn}$ and two neutrons
C $^{135}_{52}\text{Te}$ and two neutrons
D $^{133}_{52}\text{Te}$ and three neutrons

Answer: C

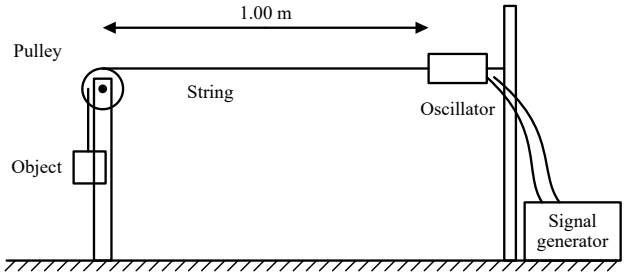
$$^{235}_{92}\text{U} + {}^{1}_{0}\text{n} \rightarrow {}^{99}_{40}\text{Zr} + {}^{A}_{Z}\text{X} + k\,{}^{1}_{0}\text{n}$$

$Z = 92 - 40 = 52$ (Te); $A + k = 236 - 99 = 137 \Longrightarrow {}^{135}_{52}\text{Te}$ and two neutrons

Explanation: Conserve the nucleon number and the proton number. Only option C balances both.

Section B [15 marks]

16. One end of a string is attached to an oscillator that oscillates at frequency 120 Hz. The other end of the string passes over a pulley that is attached to an object as shown in the diagram below.



A standing wave with four loops is formed in the string when a certain mass, M, is attached to the object. The mass per unit length of the string is 1.00×10^{-4} kg m^{-1}.

(a) Explain the formation of nodes in the standing wave. [2]

Answer

The wave from the oscillator is reflected at the pulley, and the incident and reflected waves (same frequency and amplitude, opposite directions) superpose — 1

At the nodes the two waves are always in antiphase, so they interfere destructively and the displacement is always zero — 1

2

Explanation: The ends of the string (oscillator and pulley) are nodes.

(b) Determine the speed of the transverse wave in the string. [3]

Answer

Four loops in 1.00 m: $L = 4 \times \frac{\lambda}{2}$ — 1

$\lambda = 0.500$ m — 1

$v = f\lambda = (120)(0.500) = 60.0$ m s^{-1} — 1

3

Explanation: Each loop of a standing wave is half a wavelength long.

(c) Calculate the value of M. [3]

Answer

$v = \sqrt{(\frac{T}{\mu})} \Longrightarrow T = \mu v^2$ — 1

$T = (1.00 \times 10^{-4})(60.0)^2 = 0.360$ N — 1

$M = \frac{T}{g} = \frac{0.360}{9.81} = 0.0367$ kg — 1

3

Total for question **8**

Explanation: The tension in the string is the weight of the hanging object.

17. The light of wavelength 560 nm is incident normally on a diffraction grating of 300 lines per mm.

(a) Determine the angle for the second order diffraction. [3]

Answer

$d = \dfrac{1}{300 \times 10^3} = 3.33 \times 10^{-6}$ m — 1

$d \sin\theta = n\lambda \Longrightarrow \sin\theta = \dfrac{2(560 \times 10^{-9})}{3.33 \times 10^{-6}} = 0.336$ — 1

$\theta = 19.6°$ — 1

3

Explanation: 300 lines per mm = 3.00×10^5 lines per metre.

(b) Determine the highest order for the diffraction pattern. [2]

Answer

$\sin\theta \leq 1 \Longrightarrow n \leq \dfrac{d}{\lambda} = \dfrac{3.33 \times 10^{-6}}{560 \times 10^{-9}} = 5.95$ — 1

Highest order $n = 5$ — 1

2

Explanation: Round down: the order must be a whole number with sin θ less than 1.

(c) Why is the diffraction pattern **not** observed if the wavelength is 5000 nm? Explain your answer. [2]

Answer

$d/\lambda = \dfrac{3.33 \times 10^{-6}}{5000 \times 10^{-9}} = 0.67 < 1$ — 1

Even for $n = 1$, $\sin\theta = \lambda/d$ would exceed 1, so no diffraction order can form — 1

2

Total for question **7**

Explanation: The wavelength is larger than the grating spacing, so only the zero-order (straight-through) beam exists.

Section C [30 marks]

18. (a) What is meant by *simple harmonic motion*? [2]

Answer

The acceleration is directly proportional to the displacement from a fixed point (the equilibrium position) — 1

and is always directed towards that point ($a = -\omega^2 x$) — 1

2

Explanation: Both parts are needed: proportionality and the restoring direction.

(b) A needle of a sewing machine moves vertically up and down in a simple harmonic motion. The needle and its holder have a total mass of 30.0 g. The needle tip rises up to 15.0 mm above the surface of the table of the machine and descends to 25.0 mm below the surface of the table of the machine. The frequency of the motion is 14.0 Hz.

(i) Deduce the equation for motions of the needle. [4]

Answer

Amplitude $A = \frac{15.0 + 25.0}{2} = 20.0 \text{ mm} = 2.00 \times 10^{-2} \text{ m}$ — 1

Equilibrium position is $25.0 - 20.0 = 5.00$ mm below the table surface — 1

$\omega = 2\pi f = 2\pi(14.0) = 88.0 \text{ rad s}^{-1}$ — 1

$x = 2.00 \times 10^{-2} \sin(88.0t)$ m, measured from the equilibrium position 5.00 mm below the table — 1

4

Explanation: The motion is symmetrical about the equilibrium position, which is midway between the highest and lowest points, not at the table surface.

(ii) Determine the maximum speed and the position of the needle tip with respect to the surface of the table of the machine. [3]

Answer

$v_{max} = A\omega = (2.00 \times 10^{-2})(88.0)$ — 1

$v_{max} = 1.76 \text{ m s}^{-1}$ — 1

at the equilibrium position, 5.00 mm below the table surface — 1

3

Explanation: The speed is greatest where the displacement from equilibrium is zero.

(iii) Determine the contact force that acts on the cloth on the surface of the table of the sewing machine during sewing. [3]

Answer

At the table surface the needle is 5.00 mm above the equilibrium: $a = \omega^2 x = (88.0)^2(5.00 \times 10^{-3})$ — 1

$a = 38.7 \text{ m s}^{-2}$ — 1

$F = ma = (0.0300)(38.7) = 1.16 \text{ N}$ — 1

3

Explanation: The acceleration depends on the displacement from the equilibrium position, not from the table surface.

(c) Differentiate between free oscillation and forced oscillation. [3]

Answer

Free oscillation: the system oscillates at its own natural frequency after an initial disturbance, with no periodic external force — 1

Forced oscillation: a periodic external (driving) force acts; the system oscillates at the frequency of the driving force — 1

In forced oscillation the amplitude depends on the driving frequency and is greatest at resonance (when it equals the natural frequency) — 1

3

Total for question **15**

Explanation: A free oscillation without damping keeps constant amplitude; a forced oscillation reaches a steady amplitude set by the driver.

19. (a) A spherical mirror has a radius of curvature 30.0 cm. When an object is placed in front of the mirror, an upright image is formed 18.0 cm from the mirror.

(i) Deduce the type of the spherical mirror. [3]

Answer

$f = \frac{r}{2} = 15.0 \text{ cm}$ — 1

A convex mirror always forms an upright virtual image within its focal length (less than 15.0 cm from the mirror) — 1

The upright image is 18.0 cm from the mirror (beyond f), so the mirror is concave (object inside f) — 1

3

Explanation: A concave mirror gives an upright, virtual, magnified image when the object is between the mirror and the focal point.

(ii) Determine the distance of the object from the mirror. [4]

Answer

$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$, virtual image: $v = -18.0$ cm — 1

$\frac{1}{u} = \frac{1}{15.0} + \frac{1}{18.0}$ — 1

$= 0.122\ \text{cm}^{-1}$ — 1

$u = 8.18$ cm — 1

4

Explanation: Use the real-is-positive convention: a virtual image has a negative image distance.

(b) A black dot is located 6.00 cm from the surface of a glass sphere of diameter 5.00 cm with a refractive index of 1.52. Determine the position of the final image of the black dot formed by the glass sphere. [5]

Answer

First surface ($r = +2.50$ cm): $\frac{1}{6.00} + \frac{1.52}{v_1} = \frac{1.52 - 1}{2.50}$ — 1

$v_1 = 36.8$ cm (beyond the first surface) — 1

For the second surface this is a virtual object: $u_2 = -(36.8 - 5.00) = -31.8$ cm, $r = -2.50$ cm — 1

$\frac{1.52}{-31.8} + \frac{1}{v_2} = \frac{1 - 1.52}{-2.50}$ — 1

$v_2 = 3.91$ cm beyond the far surface of the sphere (real image) — 1

5

Explanation: Treat the sphere as two refracting surfaces. The image from the first surface lies beyond the second surface, so it is a virtual object for the second refraction.

(c) Compare **three** characteristics of the images formed by a convex mirror and a concave lens. [3]

Answer

Both images are virtual — 1

Both images are upright — 1

Both images are diminished (and lie within the focal length); the convex mirror image is behind the mirror, the concave lens image is on the same side as the object — 1

3

Total for question **15**

Explanation: A convex mirror and a concave lens are both diverging, so they give the same kind of image for any real object.

20. (a) Using postulates of the Bohr model of an atom, derive an expression for the radii of the orbits of a hydrogen atom. [5]

Answer

The Coulomb force provides the centripetal force: $\frac{mv^2}{r} = \frac{e^2}{4\pi\varepsilon_0 r^2}$ 1

Angular momentum is quantised: $mvr = \frac{nh}{2\pi}$, $n = 1, 2, 3, \ldots$ 1

$v = \frac{nh}{2\pi mr}$ 1

$m\frac{n^2h^2}{4\pi^2m^2r^2} = \frac{e^2}{4\pi\varepsilon_0 r}$ 1

$r = \frac{\varepsilon_0 h^2 n^2}{\pi m e^2}$ 1

5

Explanation: The radii increase as n^2; for $n = 1$ this gives the Bohr radius, 5.3×10^{-11} m.

(b) A beam of electrons with electron current 10.0 mA is directed to a metal target in an X-ray tube that operates at 140 kV with an efficiency of 1.0%.

(i) Determine the number of electrons that hit the target in every second. [3]

Answer

$I = Ne \Longrightarrow N = \frac{I}{e}$ 1

$N = \frac{10.0 \times 10^{-3}}{1.60 \times 10^{-19}}$ 1

$N = 6.25 \times 10^{16}\ \text{s}^{-1}$ 1

3

Explanation: Current is the charge passing per second.

(ii) Determine the minimum wavelength of the X-rays emitted. [2]

Answer

$eV = \frac{hc}{\lambda_{\text{min}}} \Longrightarrow \lambda_{\text{min}} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{(1.60 \times 10^{-19})(140 \times 10^3)}$ 1

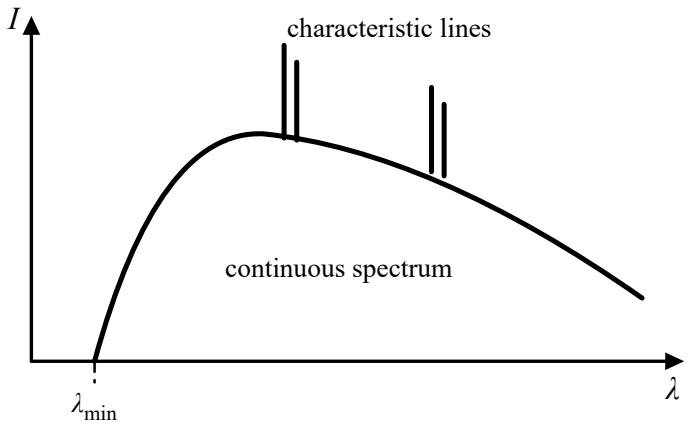
$\lambda_{\text{min}} = 8.88 \times 10^{-12}$ m 1

2

Explanation: The minimum wavelength comes from an electron that loses all its kinetic energy in a single collision, producing one photon.

(iii) Sketch and label a graph of intensity, I, against wavelength, λ, of the expected X-ray spectrum from the X-ray tube. [2]

Answer



Continuous spectrum starting at λ_{min}	1
Sharp characteristic lines superimposed on the continuous spectrum	1
	2

Explanation: The continuous spectrum comes from decelerating electrons (bremsstrahlung); the lines come from electron transitions in the target atoms.

(iv) Determine the average energy per second of the X-rays produced. [3]

Answer

Electrical power supplied = $IV = (10.0 \times 10^{-3})(140 \times 10^{3}) = 1400$ W	1
X-ray power = 1.0% of 1400 W	1
= 14 W	1
	3
Total for question	**15**

Explanation: The other 99% of the energy becomes heat in the target, which is why the anode must be cooled.